

Extended Syllabus

Course Title	Computational Physics 2	Semester	2026 Fall
Credit	3	Course Number	PHY4008-01
Class Time	Tue, Thu 13:30-14:45	Enrollment Eligibility	Physics Major (Juniors and Seniors)

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I. Course Overview

1. Description

This course develops practical computational skills for modern physics research, including numerical differential equations, eigenvalue problems, electronic band structures, Monte Carlo methods, density functional theory, parallel computing, and quantum many-body systems. Emphasis is placed on reliable implementation, numerical validation, reproducibility, physical interpretation, and responsible use of AI-assisted programming tools. Each student develops one individual project from a working midterm prototype to a completed final study.

2. Prerequisites

- Essential: Quantum Mechanics, Basic Python Programming
- Optional: Statistical Mechanics, Solid State Physics

3. Course Format (%)

Lecture	Discussion	Experiment / Practicum	Field study	Presentations	Other
35 %	10 %	45 %	%	10 %	%

4. Evaluation (%)

Mid-term Exam	Final exam	Quizzes	Presentations	Projects	Assignments	Participation	Other
0 %	0 %	0 %	80 %	0 %	0 %	20 %	%

II. Course Objectives

- Implement and validate computational methods for solving physics problems.
- Develop one computational research project from a working prototype to a reproducible final study.
- Use AI-assisted tools responsibly by testing and explaining AI-assisted code.

III. Course Format

- **Lectures and Coding Practice**
The course consists of two 75-minute sessions per week, combining physical concepts, numerical methods, hands-on coding, and project development. Students must bring their own laptops to every class.
- **Semester Project**
Each student develops one individual computational physics project throughout the semester. The midterm and final presentations must address the same project unless a major topic change is approved in advance.
- **Midterm and Final Project Presentations**
Students give two in-class presentations: a working prototype in Week 7 and a completed project in Week 15. Required materials include slides, executable code, and a README. There are no classes during the midterm and final examination weeks.

IV. Course Requirements and Grading Criteria

- **Course Evaluation**
 - Midterm Project Presentation (30%): Working prototype, preliminary results, slides, source code, README, presentation, and Q&A.
 - Final Project Presentation (50%): Completed project, validated results, reproducible code, concise report, presentation, and Q&A.
 - Participation (20%): Attendance, in-class coding, project checkpoints, questions, and constructive peer feedback.
- **Presentation Evaluation Criteria**
 - Problem Definition and Physical Significance
 - Technical Accuracy and Numerical Validation
 - Code Implementation and Reproducibility
 - Scientific Interpretation, Delivery, and Q&A

V. Course Policies

1. **Attendance and Participation**
Attendance is mandatory. Students with 5 or more absences will receive an F grade. Participation also includes in-class coding, project checkpoints, questions, and peer feedback.
2. **Project and Presentation Deadlines**
All project materials must be submitted by the announced deadlines. Late submissions without prior approval may be penalized. Submitted code must run using the instructions in the README.
3. **Academic Integrity**
All submitted work must be your own. Plagiarism is strictly prohibited and may result in disciplinary action.
4. **Use of AI Tools (e.g., GitHub Copilot, ChatGPT)**
AI tools are allowed, but students must understand, test, and take responsibility for all code. Substantial AI assistance must be disclosed in the README. Students must be able to explain and validate every submitted result.
5. **Collaboration**
Peer discussion is encouraged, but each student must maintain an individual project repository and submit independently unless otherwise specified.

VI. Materials and References

1. **Course Materials**
 - Weekly lecture notes and Jupyter notebooks
 - Sample codes, project templates, and reproducibility checklists
2. **Tools and Platforms**

- **Visual Studio Code (VSCode)** – recommended coding environment
- **GitHub Education Pack** – access to Copilot and other student developer tools (<https://education.github.com/pack>)
- **Google Colab** – optional for cloud-based coding

3. Resources for AI-Assisted Programming

- GitHub Copilot official documentation (<https://docs.github.com/en/copilot>)
- AI-assisted coding, verification, and disclosure guidelines
- Class examples of testing and validating AI-assisted code

VII. Course Schedule

(* Subject to change)

Week 1	Learning Objectives	Reproducible computational workflow
	Topics	Course introduction; Python/Jupyter refresh; Git and GitHub; responsible use of AI tools
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	N/A
Week 2	Learning Objectives	Numerical verification and error analysis
	Topics	Floating-point error, convergence, testing, and scientific visualization
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	N/A
Week 3	Learning Objectives	Numerical differential equations
	Topics	ODE integrators, stability, and physical applications
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks

	Assignments	Project topic exploration
Week 4	Learning Objectives	Numerical eigenvalue problems
	Topics	Bound states, 1D wells, and matrix diagonalization
	Class Work (Methods)	Theory & hands-on (9/22); no class on 9/24 (Chuseok)
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	9/29: One-page project proposal due
Week 5	Learning Objectives	Matrix methods
	Topics	Tight-binding models and electronic band structures
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	Project implementation
Week 6	Learning Objectives	Monte Carlo methods
	Topics	Metropolis algorithm, Ising model, and high-dimensional integration
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	10/8: Working prototype checkpoint
Week 7	Learning Objectives	Midterm Project Presentations
	Topics	Working prototype, preliminary results, and validation plan
	Class Work (Methods)	In-class presentations and Q&A
	Materials (Required Readings)	Slides, code, and project repository

	Assignments	10/13 & 10/15: Midterm Project Presentations
Week 8	Learning Objectives	No class (Midterm Examination Week; no written midterm examination)
	Topics	
	Class Work (Methods)	
	Materials (Required Readings)	
	Assignments	
Week 9	Learning Objectives	Density Functional Theory for Atoms (I)
	Topics	DFT basics, Kohn-Sham equations, SCF loop, and Poisson solver
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	10/29: Feedback response and revision plan due
Week 10	Learning Objectives	Density Functional Theory for Atoms (II)
	Topics	SCF convergence, total-energy functionals, and numerical mixing
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	Project development
Week 11	Learning Objectives	Foundations of parallel computing
	Topics	Profiling, vectorization, multiprocessing, and scaling
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks

	Assignments	Project development
Week 12	Learning Objectives	MPI and parallel numerical methods
	Topics	Message passing, collective operations, and parallel PDE calculations
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	11/19: Code and reproducibility review
Week 13	Learning Objectives	Quantum Spin Systems (I)
	Topics	Spin-chain models, basis construction, and Hamiltonian matrices
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	Project development
Week 14	Learning Objectives	Quantum Spin Systems (II)
	Topics	Sparse matrices, exact diagonalization, and selected many-body problems
	Class Work (Methods)	Theory & hands-on
	Materials (Required Readings)	Lecture notes & Jupyter notebooks
	Assignments	12/7: Final project materials due
Week 15	Learning Objectives	Final Project Presentations
	Topics	Completed computational study, validated results, and physical interpretation
	Class Work (Methods)	In-class presentations and Q&A
	Materials (Required Readings)	Slides, code, project repository, and concise report

	Assignments	12/8 & 12/10: Final Project Presentations
Week 16	Learning Objectives	No class (Final Examination Week; no written final examination)
	Topics	
	Class Work (Methods)	
	Materials (Required Readings)	
	Assignments	

VIII. Special Accommodations

Students who require accommodations due to disabilities or other reasons are encouraged to contact the course instructor.

IX. Aid for the Challenged Students

For students with disabilities, accommodations can be provided for assignments and evaluations. Please consult with the course instructor in advance.